Rheinisch-Westfälische Technische Hochschule Aachen
Alessandro Berti

Project Initialization: Implementation of "Conformance Predictions"

Group 4:
Richard Schuppener, Cem Colak

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1 Introduction

This document encompasses the set-up phase of Cem and Richards project "Conformance Predictions" supervised by Dr. Alessandro Berti. The scope of this project is located in the field of Process Mining and Data Science [1]. We aim to provide a fully functional Python (+PM4Py) based, modular repository that integrates **Process Mining** and **Machine Learning** to retrieve relevant insights. Of course, the projects output shall be applicable to real-world data in form of event logs and normative process models. In the following, exemplary business cases for our projects output are presented, the theoretical and technical feasibility is analyzed and a project charter is presented. Finally we provide personal descriptions of the teams members (Richard & Cem) and both assign roles and provide a personal phase review.

2 Business Case

2.1 Client

Insurance company F is a internationally established insurer for road vehicles. It handles processes such as *application-to-approval* and *issue-to-resolution*. In the following we are going to focus on established clients of company F that submit an insurance claim as in the issue-to-resolution process. There are multiple stages of this process where conformance deviations may take place, such as the claim registration (sub-) process: "After a claim is registered, it is examined by a claims officer who then writes a settlement recommendation. This recommendation is then checked by a senior claims officer who may mark the claim as "OK" or "Not OK". If the claim is marked as "Not OK", it is sent back to the claims officer and the recommendation is repeated. If the claim is "OK", the claim handling process proceeds." [2]. We modeled this processa as BPMN diagram in 1

2.2 Example Use Case 1:

Company F notices that the claim handling regularly fails because of issues that have been introduced in the claim registration process which it directly proceeds. Such issues include:

1. The activity "Evaluate severity" was skipped and thus the activity "Send forms" is executed only with the basic forms sent to the customer and not the (in severe cases required) additional forms.

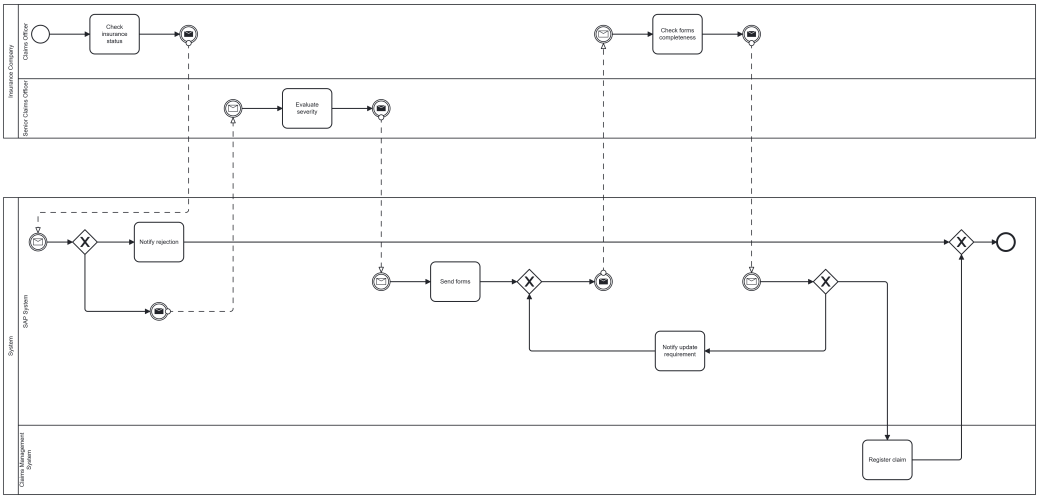


Figure 1: Claim registration process for company F.

2. The activity "Check forms completeness" was emitted and thus the claim is registered even though not all required forms are present.

It is self-evident how these deviations from standard procedure cause problems later on in the claim handling process. Depending on the exact conformance violation in the claim registration process, deviations in the subsequent process may or may not arise. To get a basis on which to act on, company F hopes to predict conformance deviations that may occur in subsequent parts of the process by deploying a ML-driven conformance prediction strategy.

2.3 Example Use Case 2:

Company F also suspects that a variety of case attributes, including the claimant's age, sex, and the total claim amount might be useful features for predicting the eventual trajectory of the case. For instance, a claim with a high monetary value filed by a claimant in a specific demographic might be predicted to have a higher probability of being marked as "Not OK" during the senior officer's review, or it might be flagged as being at risk for a conformance deviation, such as the omission of a mandatory secondary verification. By analyzing these attributes early on, company F can anticipate whether a case will follow the standard workflow or if it is likely to encounter deviations, allowing for proactive management and more accurate outcome predictions.
A second sub-process follows this initial registration: the *Claim Settlement*

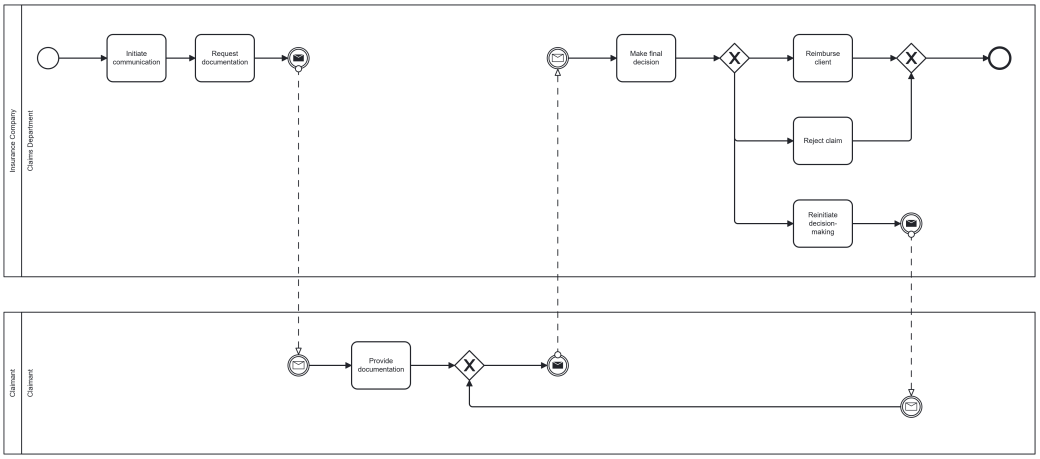


Figure 2: Claim communication and settlement process for company F.

and Communication phase. "In this stage, the claims officer initiates direct communication with the claimant to request additional documentation or clarify the circumstances of the incident. Based on the gathered information, a final decision is made to either reimburse the client, reject the claim, or reinitiate the decision-making process if new evidence suggests a different path." 2 (this process is not taken from the book "Fundamentals of Business Process Management").

Conformance deviations are have been recorded during this interaction; for example, the officer might bypass the mandatory "Request Information" step and move directly to a rejection, or they might fail to provide the required justification when reinitiating the process. These deviations are candidates for prediction, as early indicators in the claim registration and especially case attributes often exhibit a higher likelihood of non-compliant behavior during these final decision steps.

3 Feasibility Study

3.1 Theoretical Feasibility

3.2 Technical Feasibility

4 Project Charter

The milestones and their respective deadlines for submission of the required materials are:

- Project Setup:
- Requirements Engineering:
- Sprint 1:
- Sprint 2:
- Sprint 3:
- Testing and Assessment:

It is essential to acknowledge that the schedule presented here is preliminary and may be subject to adjustments. This flexibility accommodates unforeseen personal or professional circumstances that may arise during the project's execution, allowing team members to navigate challenges effectively.

5 Tools

5.1 Trello

In order to plan, organize, and track the development of our work, we rely on a project management tool called Trello. It is based on the concept of boards, lists, and cards, allowing us to break down complex tasks into smaller ones and distribute them among team members. We utilize features such as due dates and labels to further customize our tasks and distinguish them from one another. This web-based platform makes it easy for group members to monitor the progress of each task and identify their responsibilities. In conclusion, Trello's user-friendly interface and comprehensive features significantly contribute to the success of our project.

Link to our Trello board: *Group 4 Trello*

5.2 Git and GitLab

Version control systems are indispensable for modern software development, especially in collaborative environments like academic projects. For our PM4PY based project we used Git and GitLab, which will use the following:

Git, at its core, is a distributed version control system that tracks changes in source code over time. Its key strengths lie in its ability to create branches for remote independent development, merge code easily, and revert to previous versions. This allows team members to work concurrently on different

features or bug fixes without interfering with each other's progress. Changes are transparently tracked and loss of progress is mitigated. Remote repositories enable the developing team to get the most up-to-date version of the project and develop from there.

GitLab on the other hand, provides a web-based platform built around Git, extending its functionalities with a suite of project management and collaboration tools. These include issue tracking, code review, continuous integration/continuous delivery (CI/CD) pipelines for example to automatically run tests, and wiki pages for documentation. Additionally, analytics are automatically created and on-demand displayed such as commits per member, time of commits, code review analytics, CI/CD analytics and many more. By integrating these features, GitLab creates a centralized hub for project activity.

5.3 PM4Py

PM4Py (Process Mining for Python) [3] is an library that was created and over many years developed the Frauenhofer Institute (FiT), but development was handed off to its spin-off *Process Intelligence Solutions* (P. I. S.). The library is open-source and encompasses a plethora of state-of-the-art process mining algorithms for filtering, discovery, conformance checking, visualization etc. In regards to our project and the overall topic of Conformance Predictions, PM4Py supports f.e. token-based, alignment-based and footprint-based conformance diagnostics, while also offering many visualization techniques. Special methods for filtering and extracting features from event logs is also supported, which makes PM4Py it perfect for our project.

6 Team

6.1 Personal Description

6.1.1 Richard Schuppener

Richard did his Bachelors degree in Computer Science from 2022-2025 and recently started his Masters degree in Data Science (both at RWTH Aachen University). From the third semester of his bachelor on, he developed a keen interest in Process Mining - his interest in Data Science was present long

before. Currently, he is working at Fraunhofer FiT as a student assistant and previously enrolled in the following (relevant) courses:

- *Business Process Modeling and Computation*
- *Fundamentals of Business Process Management*
- *Business Process Intelligence*
- *Elements of Machine Learning and Data Science*
- *Introduction to Data Science*
- *Advanced Process Mining*
- *SPP: Conformance Checking in Python*

This concludes that he has strong theoretical backgrounds in the areas of process mining (especially conformance checking) and basics in machine learning. He is looking forward to working with Cem and honing his software development skills (in particular best practices), getting more insights into the combination of process mining and classical machine learning and ultimately delivering a complete and polished product that actually produces valuable insights.

6.1.2 Cem Colak

7 Roles

Scrum Master: We decided that the role of the scrum master will not be assigned to any of us, since our team only consists of two participants and we deem a scrum master as a single responsible person to be unfitting and unnecessary.

Official Communication/Product Owner: Cem will be assigned to handle the official communication with Alessandro and notify him about our progress, potential problems, questions and organizational information.

Tools Administration: Since Cem already set up Trello and is familiar with structuring boards, we assign him to be responsible for the Tool Administration.

Theory Expert: Due to his background in process mining, we decided to

assign Richard to the role of a theory expert.

Software Expert: This role will also be assigned to Richard because of his background in Computer Science and his experience as programmer for FiT.

Test Management: We assign the role of Test Manager to Cem, as he is a great fit for ensuring the functionality and correctness of any produced code. He has previous experience with the set-up of test suites test-first driven development.

8 Phase review

8.1 Richard Schuppener

I believe that this phase went splendid and without any hiccups. Our group work started out by quickly establishing a communication channel (WhatsApp) and by meeting up in person. The organization of weekly time slot where we plan to work together on this project was easy and fits us both. I quickly established a GitLab repository and a LaTeX template for this document and Cem directly created the Trello board. We worked evenly on the "Project Setup" document and the atmosphere among the team was friendly, productive and supportive. I am looking forward to working with Cem on this interesting project.

8.2 Cem Colak

References

[1] Van Der Aalst, Wil. (2016) *Process mining: Data science in action.* Berlin, Heidelberg: Springer Berlin Heidelberg.

[2] Dumas, M., La Rosa, M., Mendling, J., & Reijers, H. A. (2013). *Fundamentals of Business Process Management.* Springer.

[3] Berti A, van Zelst S, Schuster D. *PM4Py: A process mining library for Python.* Software Impacts. 2023